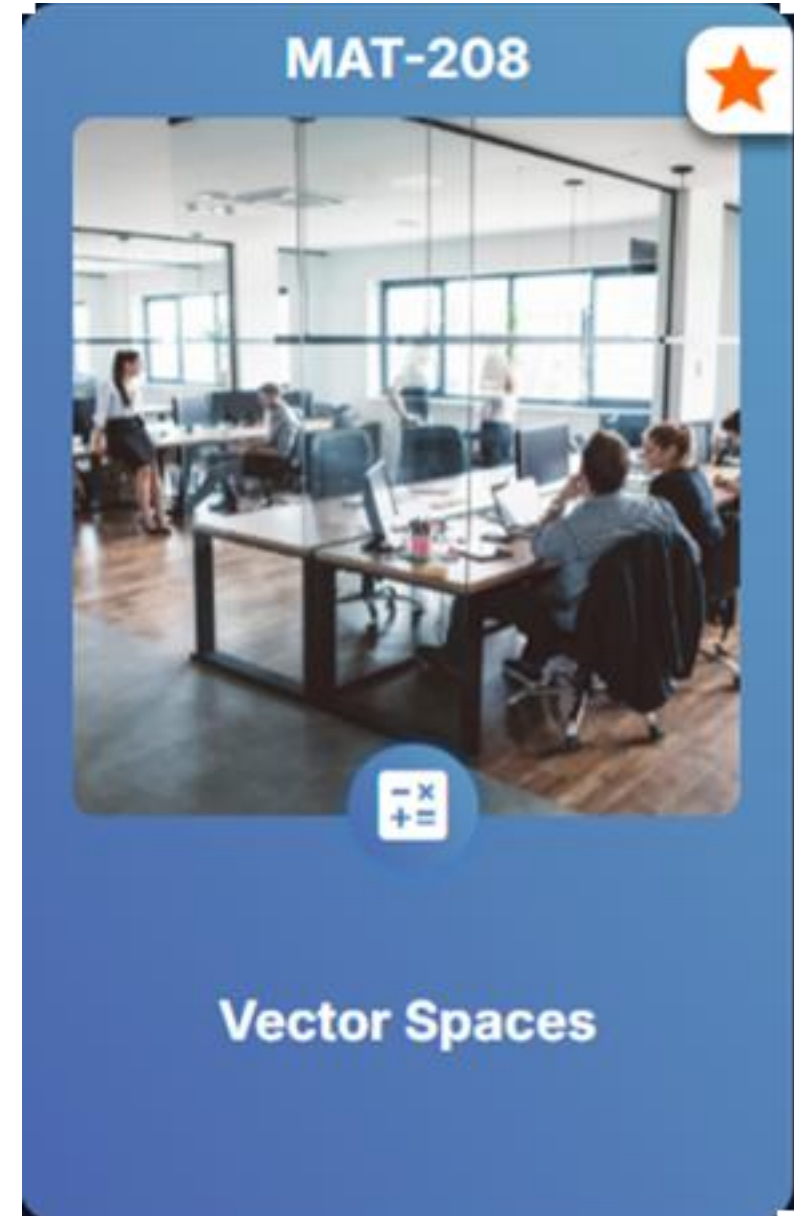


Lectures 4-5

Linear Transformation

Learning by doing



Today's Topics: Linear Transformation

- Abstract Version
- Graphical Version
- Quantum Version

ABSTRACT VERSION

Linear transformations

- V, W vector spaces with same fields F .
 - **Definition:** $T: V \rightarrow W$ s.t. $T(ca+b) = c(Ta) + Tb$ for all a, b in V . c in F . Then T is **linear**.
 - Property: $T(O) = O$. $T(ca+db) = cT(a) + dT(b)$, a, b in V , c, d in F . (equivalent to the def.)
 - **Example:** A $m \times n$ matrix over F . Define T by $Y = AX$. $T: F^n \rightarrow F^m$ is linear.
 - Proof: $T(aX+bY) = A(aX+bY) = aAX + bAY = aT(X) + bT(Y)$.

- $U: F^{1 \times m} \rightarrow F^{1 \times n}$ defined by $U(a) = aA$ is linear.
- Notation: $F^m = F^{m \times 1}$ (not like the book)
- Remark: $L(F^{m \times 1}, F^{n \times 1})$ is same as $M_{m \times n}(F)$.
 - For each $m \times n$ matrix A we define a unique linear transformation T given by $T(X) = AX$.
 - For each a linear transformation T has A such that $T(X) = AX$. We will discuss this in section 3.3.
 - Actually the two spaces are isomorphic as vector spaces.
 - If $m = n$, then compositions correspond to matrix multiplications exactly.

- Example: $T(x)=x+4$. $F=R$. $V=R$. This is not linear.
- Example: $V = \{f \text{ polynomial}: F \rightarrow F\}$
 $T:V \rightarrow V$ defined by $T(f)=Df$.

$$\bullet V = \left\{ \begin{array}{lcl} f(x) & = & c_0 + c_1x + c_2x^2 + \cdots + c_kx^k \\ Df(x) & = & c_1 + 2c_2x + \cdots + kc_kx^{k-1} \end{array} \right.$$

$$Tf(x) = \int_0^x f(t)dt$$

- **Theorem 1:** V vector space over F . basis $\alpha_1, \dots, \alpha_n$
 W another one with vectors $\beta_1, \dots, \beta_m$ (any kind
 $m \geq n$). Then **exists unique** linear transformation
 $T: V \rightarrow W$ s.t.

$$T(\alpha_j) = \beta_j, j = 1, \dots, n$$

- Proof: Check the following map is linear.

$$\begin{aligned} \alpha &= x_1 \alpha_1 + \dots + x_n \alpha_n \\ T\alpha &= x_1 \beta_1 + \dots + x_n \beta_n \end{aligned}$$

- **Null space** of $T : V \rightarrow W := \{ v \text{ in } V \mid Tv = 0 \}$.
- **Rank** $T := \dim\{Tv \mid v \text{ in } V\} \text{ in } W. = \dim \text{range } T$.
- Null space is a vector subspace of V .
- Range T is a vector subspace of W .
- Example:

$$\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

- Null space $z=t=0. X+2y=0 \dim = 1$
- Range = $W. \dim = 3$

- **Theorem:** $\text{rank } T + \text{nullity } T = \dim V$.
- **Proof:** $a_1, \dots, a_k$ basis of N . $\dim N = k$. Extend to a basis of V : $a_1, \dots, a_k, a_{k+1}, \dots, a_n$.
 - We show $T a_{k+1}, \dots, T a_n$ is a basis of R . Thus $n-k = \dim R$.
 $n-k+k=n$.
 - Spans R :
 - Independence:

$$\begin{aligned} v &= x_1 \alpha_1 + \dots + x_n \alpha_n \\ Tv &= x_{k+1} T(\alpha_{k+1}) + \dots + x_n T(\alpha_n) \end{aligned}$$

$$\begin{aligned} \sum_{i=k+1}^n c_i T \alpha_i &= 0 \\ T\left(\sum_{i=k+1}^n c_i \alpha_i\right) &= 0 \\ \sum_{i=k+1}^n c_i \alpha_i &\in N \\ \sum_{i=k+1}^n c_i \alpha_i &= \sum_{i=1}^k c_i \alpha_i \\ c_i &= 0, i = k+1, \dots, n \end{aligned}$$

- Theorem 3: A $m \times n$ matrix.
Row rank A = Column rank A .
- Proof:
 - column rank A = rank T where $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $Y=AX$. e_i goes to i -th column. So range is spanned by column vectors.
 - rank T + nullity T = n by above theorem.
 - column rank A + $\dim S$ = n where $S = \{X | AX=O\}$ is the null space.
 - $\dim S$ = n - row rank A (Ex 15 Ch. 5)
 - row rank = column rank.

- (Ex 15 Ch. 5) $A^{m \times n}$. S solution space. R r-r-e matrix
- r = number of nonzero rows of R .
- $RX=0$ $k_1 < k_2 < \dots < k_r$. $J = \{1, \dots, n\} - \{k_1, k_2, \dots, k_r\}$.

$$\begin{array}{rcl}
 x_{k_1} & + & \sum_{j=1}^{n-r} C_{1j} u_j = 0 \\
 x_{k_2} & + & \sum_{j=1}^{n-r} C_{2j} u_j = 0 \\
 & \ddots & \\
 & + & \vdots = \vdots \\
 x_{k_r} & + & \sum_{j=1}^{n-r} C_{rj} u_j = 0
 \end{array}$$

- Solution spaces parameter $u_1, \dots, u_{n-r}$.
- Or basis E_j given by setting $u_j=1$ and other 0 and $x_{k_i} = c_{ij}$.

Algebra of linear transformations

- Linear transformations can be added, and multiplied by scalars. Hence they form a vector space themselves.
- **Theorem 4:** $T, U: V \rightarrow W$ linear.
 - Define $T+U: V \rightarrow W$ by $(T+U)(a) = T(a) + U(a)$.
 - Define $cT: V \rightarrow W$ by $cT(a) = c(T(a))$.
 - Then they are linear transformations.

- **Definition:** $L(V,W)=\{T:V\rightarrow W \mid T \text{ is linear}\}$.
- **Theorem 5:** $L(V,W)$ is a finite dim vector space if so are V,W . $\dim L = \dim V \dim W$.
- **Proof:** We find a basis:
 - Define a linear transformation $V \rightarrow W$: $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\} \subset V$
 $\mathcal{B}' = \{\beta_1, \dots, \beta_m\} \subset W$
 - We show the basis:

$$E^{p,q}(\alpha_i) = \begin{cases} 0, & i \neq q \\ \beta_p, & i = q \end{cases} = \delta_{iq} \beta_p, \quad 1 \leq p \leq m, 1 \leq q \leq n$$

$$\begin{matrix} E^{1,1}, & \dots, & E^{1,n} \\ \vdots & \ddots & \vdots \\ E^{m,1}, & \dots, & E^{m,n} \end{matrix}$$

- **Spans:** $T:V \rightarrow W$.

- We show

$$T\alpha_j = \sum_{p=1}^m A_{pj}\beta_p$$

$$T = U = \sum_{p=1}^m \sum_{q=1}^n A_{pq} E^{p,q}$$

$$\begin{aligned} U(\alpha_j) &= \sum_{p=1}^m \sum_{q=1}^n A_{pq} E^{p,q}(\alpha_j) \\ &= \sum_{p=1}^m \left(\sum_{q=1}^n A_{pq} \delta_{jp} \right) \beta_p \\ &= \sum_{p=1}^m A_{pj} \beta_p = T\alpha_j, j = 1, \dots, m \\ T &= U \end{aligned}$$

- Independence

- Suppose

$$\begin{aligned}
 U &= \sum_p \sum_q A_{pq} E^{p,q} = 0 \\
 U \alpha_j &= 0 \\
 \sum_p A_{pj} \beta_p &= 0 \\
 \{\beta_p\} & \text{ independent} \\
 A_{pj} &= 0 \text{ for all } p, j
 \end{aligned}$$

- **Example:** $V = F^m$ $W = F^n$. Then

- $M_{m \times n}(F)$ is isomorphic to $L(F^m, F^n)$ as vector spaces. Both dimensions equal mn .
 - $E^{p,q}$ is the $m \times n$ matrix with 1 at (p,q) and 0 everywhere else.
 - Any matrix is a linear combinations of $E^{p,q}$.

- **Theorem.** $T:V \rightarrow W$, $U:W \rightarrow Z$.
 $UT:V \rightarrow Z$ defined by $UT(a) = U(T(a))$ is linear.
- **Definition:** Linear operator $T:V \rightarrow V$.
- $L(V,V)$ has a multiplication.
 - **Define** $T^0 = I$, $T^n = T \dots T$. n times.
 - **Example:** A $m \times n$ matrix A $p \times m$ matrix B
 T defined by $T(X) = AX$. U defined by $U(Y) = BY$. Then
 $UT(X) = BAX$. Thus
 UT is defined by BA if T is defined by A and U by B .
 - Matrix multiplication is defined to mimic composition.

- Lemma:
 - $IU=UI=U$
 - $U(T_1+T_2)=UT_1+UT_2, (T_1+T_2)U=T_1U+T_2U.$
 - $c(UT_1)=(cU)T_1=U(cT_1).$
- Remark: This make $L(V,V)$ into linear algebra (i.e., vector space with multiplications) in fact same as the matrix algebra $M_{n \times n}(F)$ if $V=F^n$ or more generally $\dim V = n$. (Example 10. P.78)

- **Example:** $V = \{f: F \rightarrow F \mid f \text{ is a polynomial}\}$.
 - $D: V \rightarrow V$ differentiation.

- $$f(x) = c_0 + c_1x + \cdots + c_nx^n$$
- $T: V \rightarrow V: Df(x) = c_1 + \cdots + nc_nx^{n-1}$
- $DT-TD = I$. We need to show $DT-TD(f) = f$ for each polynomial f .
- (QP-PQ=iħ In quantum mechanics.)

Invertible transformations

- $T:V \rightarrow W$ is **invertible** if there exists $U:W \rightarrow V$ such that $UT=I_V$ $TU=I_W$.
U is denoted by T^{-1} .
- Theorem 7: If T is linear, then T^{-1} is linear.
- Definition: $T:V \rightarrow W$ is nonsingular if $Tc=0$ implies $c=0$
 - Equivalently the null space of T is $\{0\}$.
 - T is one to one.
- **Theorem 8:** T is nonsingular iff T carries each linearly independent set to a linearly independent set.

- **Theorem 9:** V, W $\dim V = \dim W$.
 $T: V \rightarrow W$ is linear. TFAE:
 - T is invertible.
 - T is nonsingular
 - T is onto.
- **Proof:** We use $n = \dim V = \dim W$.
 $\text{rank } T + \text{nullity } T = n$.
 - (ii) iff (iii): T is nonsingular iff $\text{nullity } T = 0$ iff $\text{rank } T = n$ iff T is onto.
 - (I) $\rightarrow$ (ii): $TX=0, T^{-1}TX=0, X=0$.
 - (ii) $\rightarrow$ (i): T is nonsingular. T is onto. T is 1-1 onto. The inverse function exists and is linear. T^{-1} exists.

Groups

- A **group** $(G, .)$:
 - A set G and an operation $G \times G \rightarrow G$:
 - $x(yz) = (xy)z$
 - There exists e s.t. $xe = ex = x$
 - To each x , there exists x^{-1} s.t. $xx^{-1} = e$ and $x^{-1}x = e$.
- Example: The set of all 1-1 maps of $\{1, 2, \dots, n\}$ to itself.
- Example: The set of nonsingular maps $GL(V, V)$ forms a group.

Isomorphisms

- V, W $T: V \rightarrow W$ one-to-one and onto (invertible). Then T is an **isomorphism**. V, W are **isomorphic**.
- Isomorphic relation is an equivalence relation: $V \sim V$, $V \sim W \Leftrightarrow W \sim V$, $V \sim W, W \sim U \Rightarrow V \sim U$.

- **Theorem 10:** Every n -dim vector space over F is isomorphic to F^n . (noncanonical)
 - Proof: V n -dimensional
 - Let $B=\{a_1, \dots, a_n\}$ be a basis.
 - Define $T:V \rightarrow F^n$ by
 - One-to-one
 - Onto
- $$\alpha = x_1\alpha_1 + \dots + x_n\alpha_n \mapsto (x_1, \dots, x_n) \in F^n$$

- Example: isomorphisms

$$F^n = \{(x_1, \dots, x_n) \mid x_i \in F\}$$

$$\cong \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mid x_i \in F \right\}$$

$$P^n(F) = \{f : F \rightarrow F \mid f(x) = c_0 + c_1x + \dots + c_nx^n\}$$

$$\cong F^{n+1}$$

Basis $\{1, x, x^2, \dots, x^n\}$

$$c_0 + c_1x + \dots + c_nx^n \mapsto (c_0, c_1, \dots, c_n)$$

There will be advantages in looking this way!

GRAPHICAL VERSION

Matrices

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \quad C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix}$$

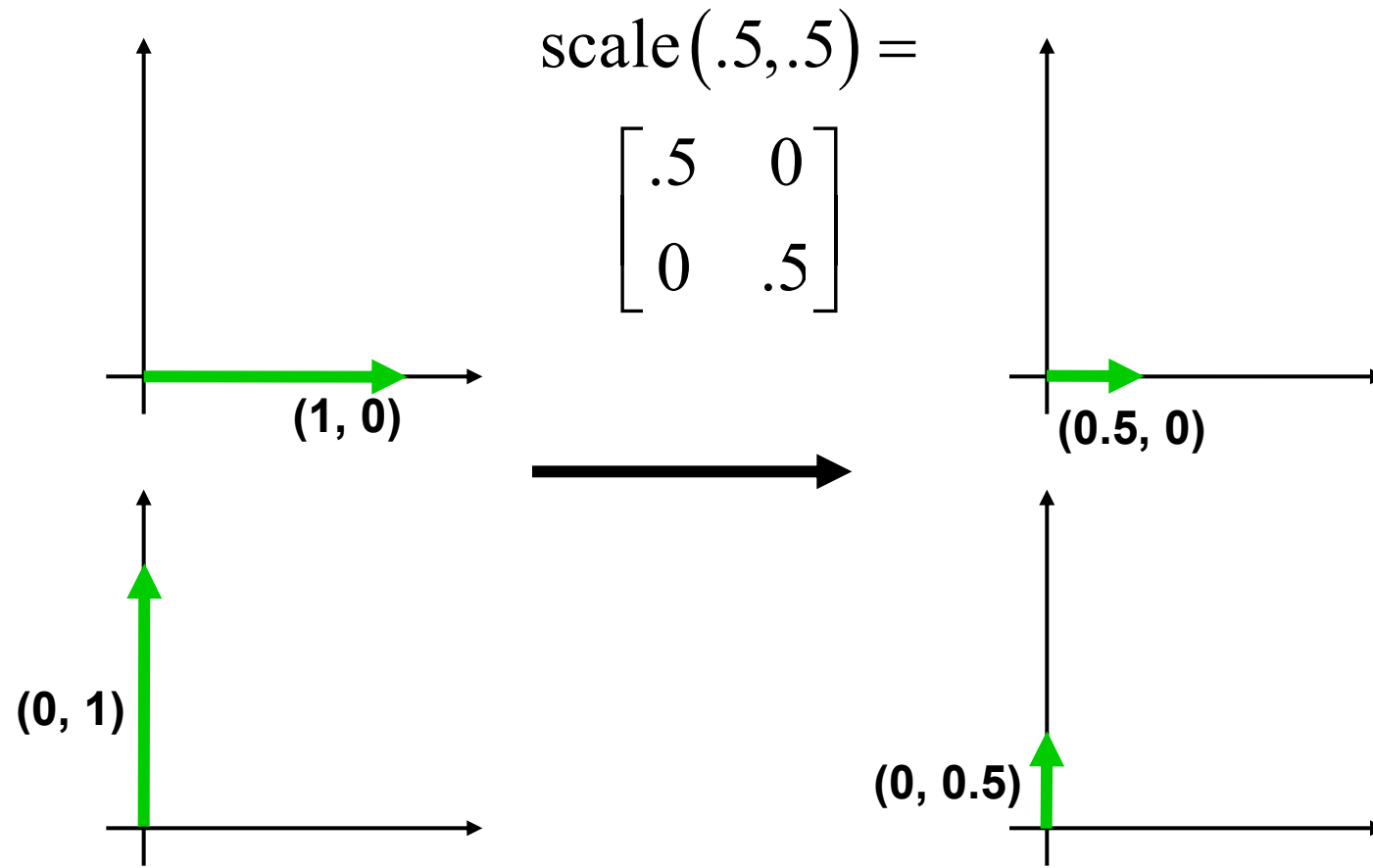
- We use 2x2, 3x3, and 4x4 matrices in computer graphics.
- We'll start with a review of 2D matrices and transformations.

Basic 2D Linear Transforms

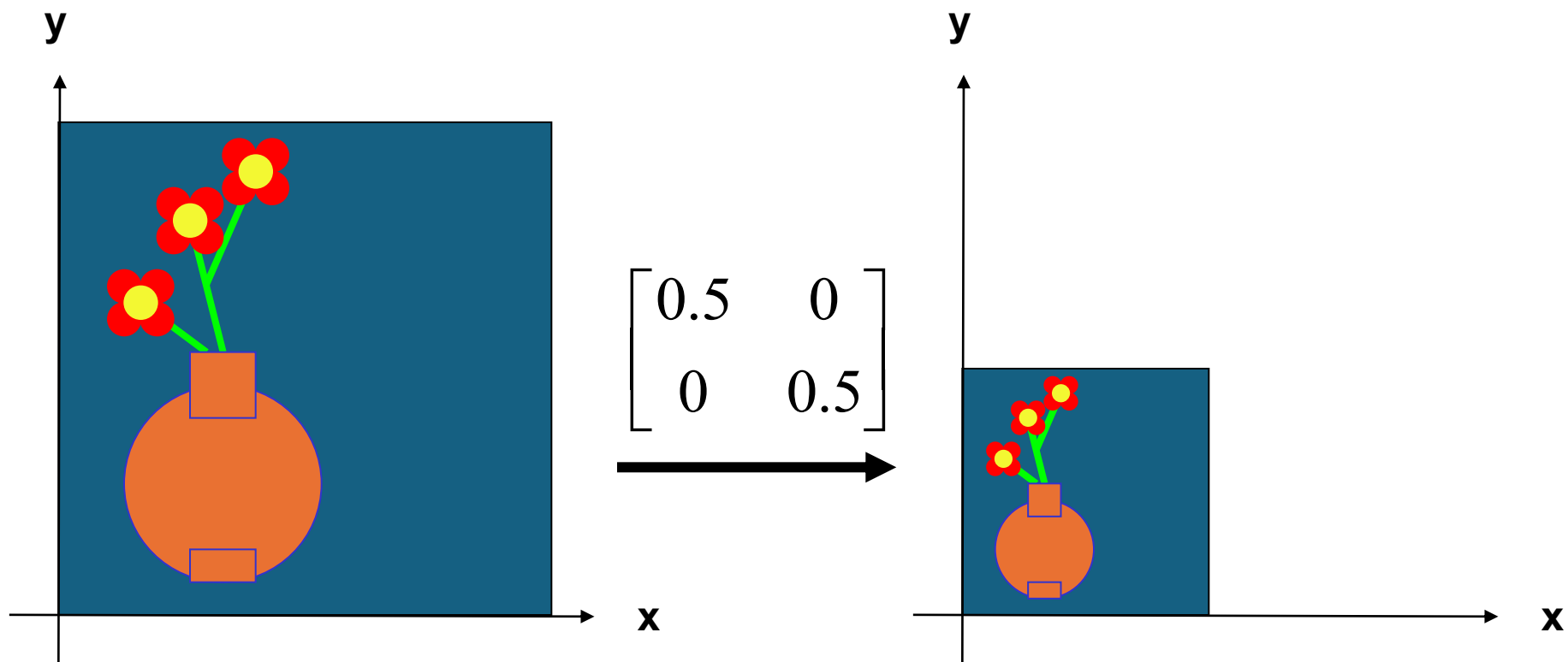
$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$$

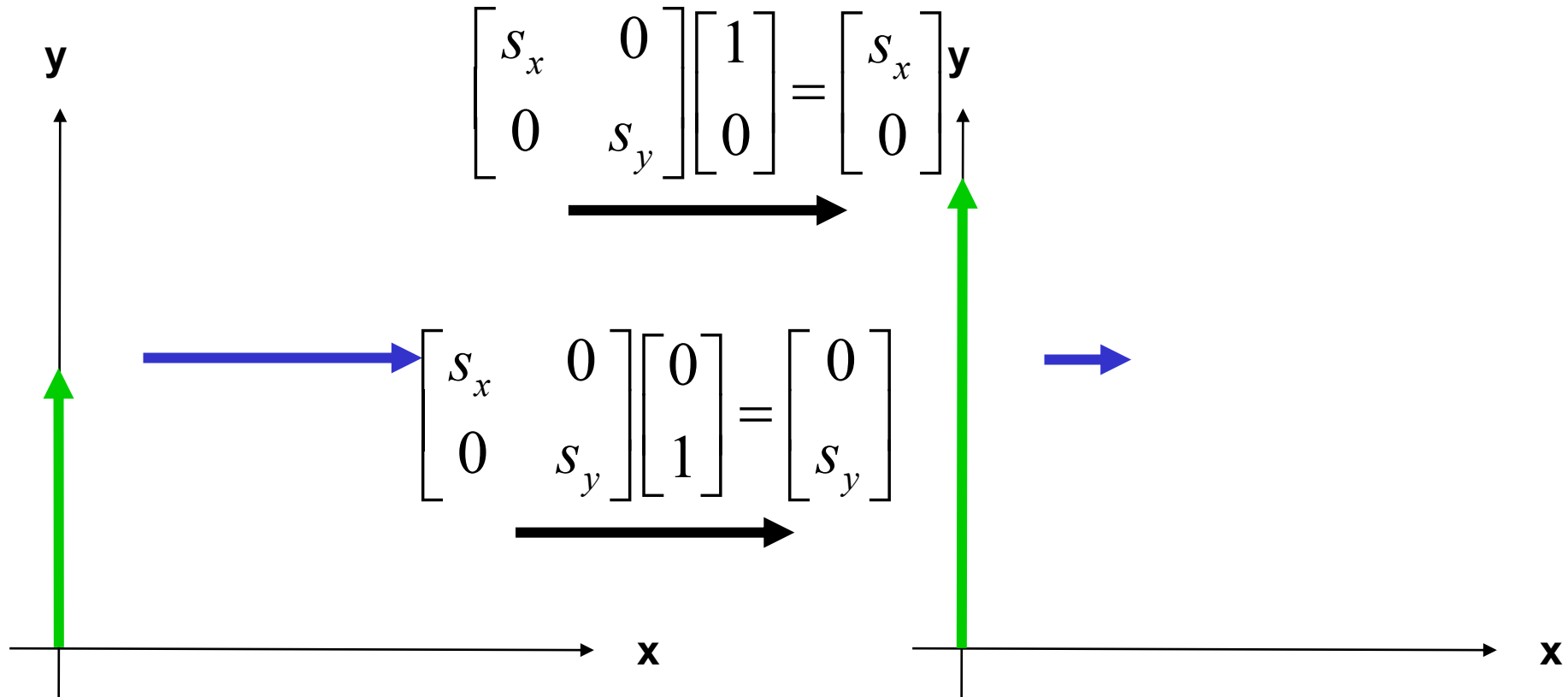
Scale by .5



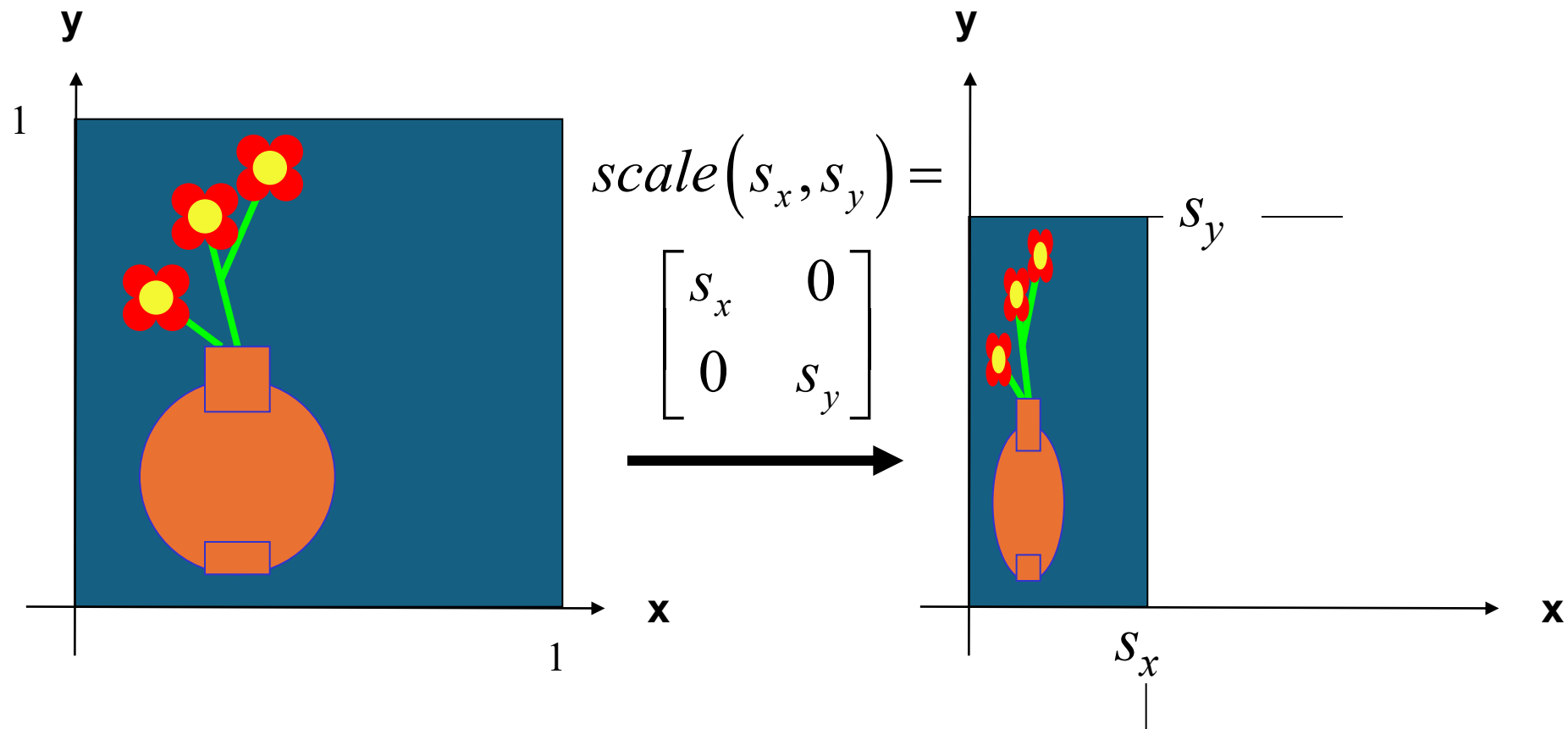
Scaling by .5



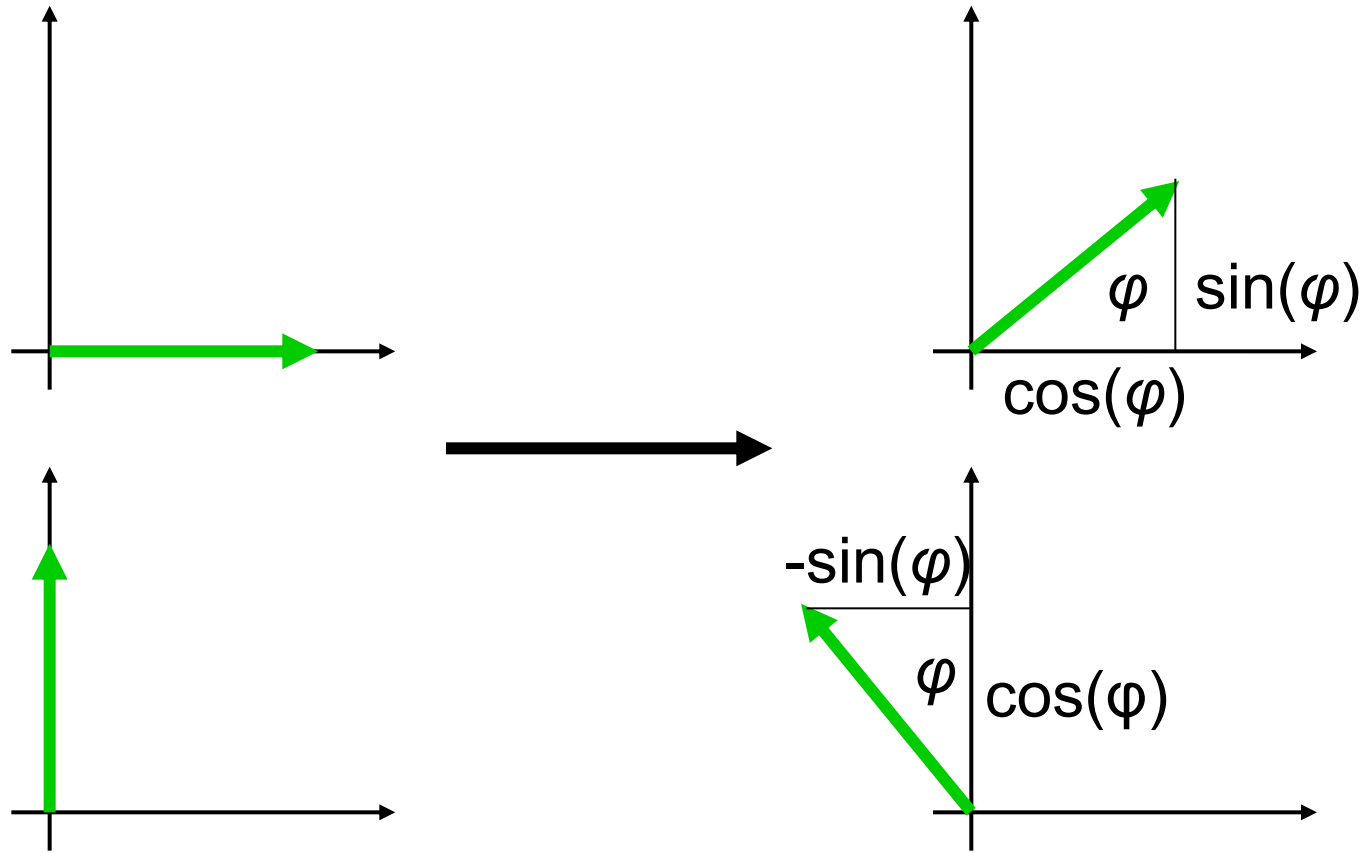
General Scaling



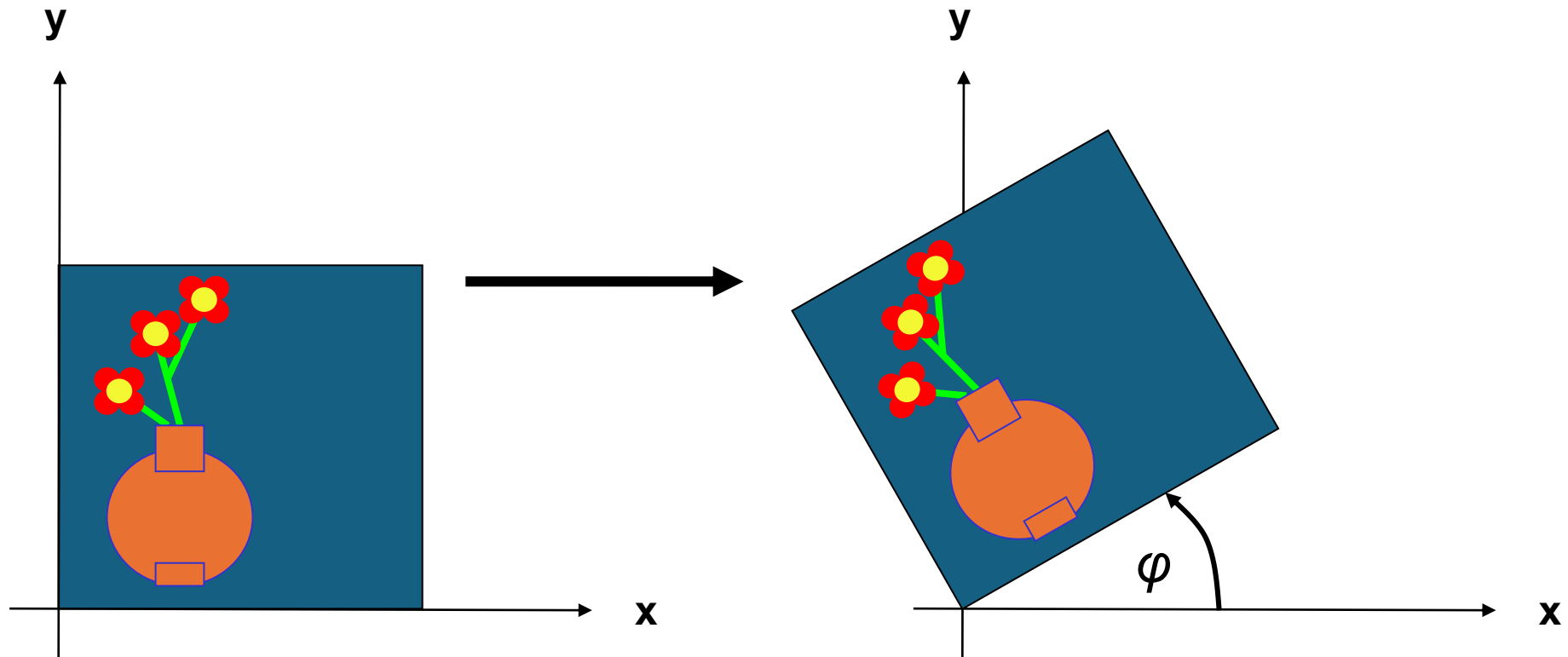
General Scaling



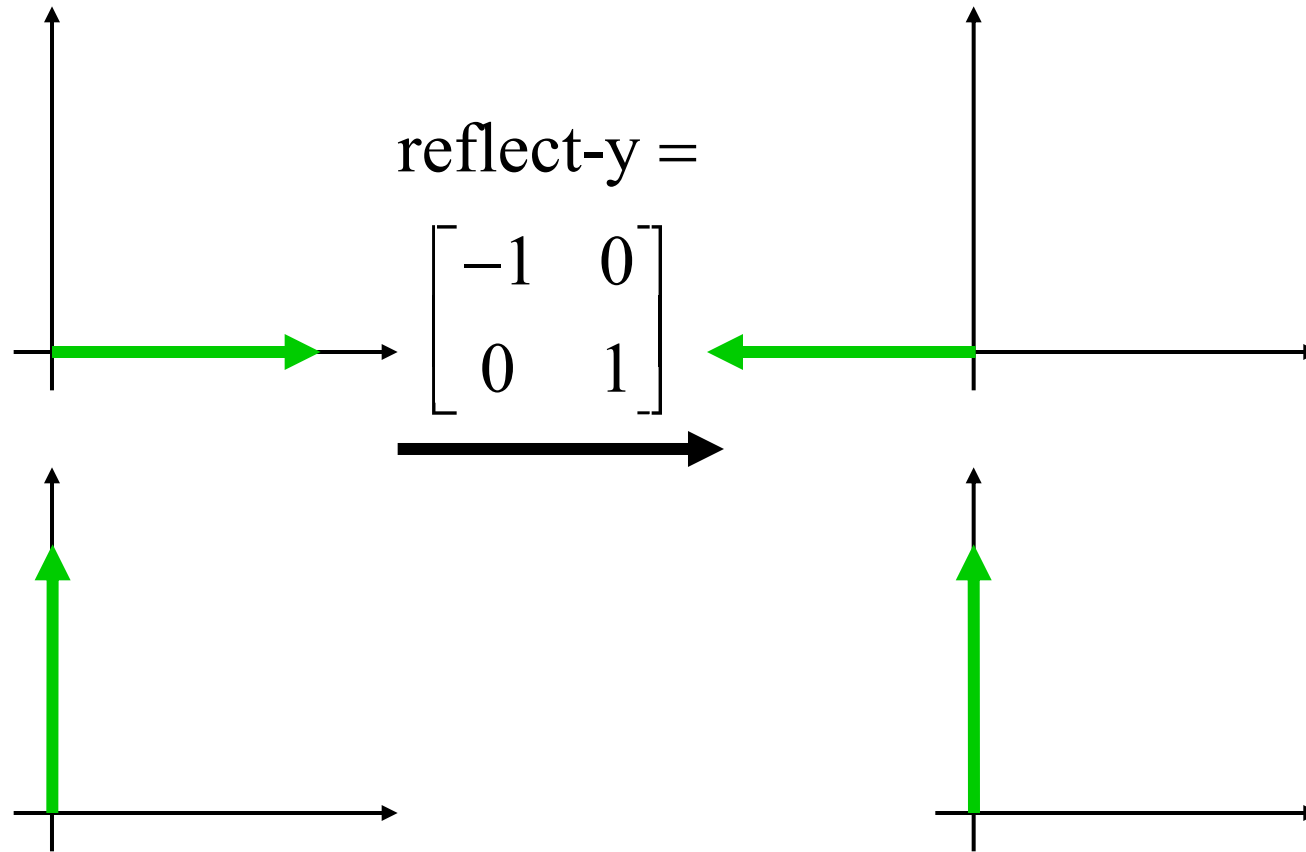
Rotation



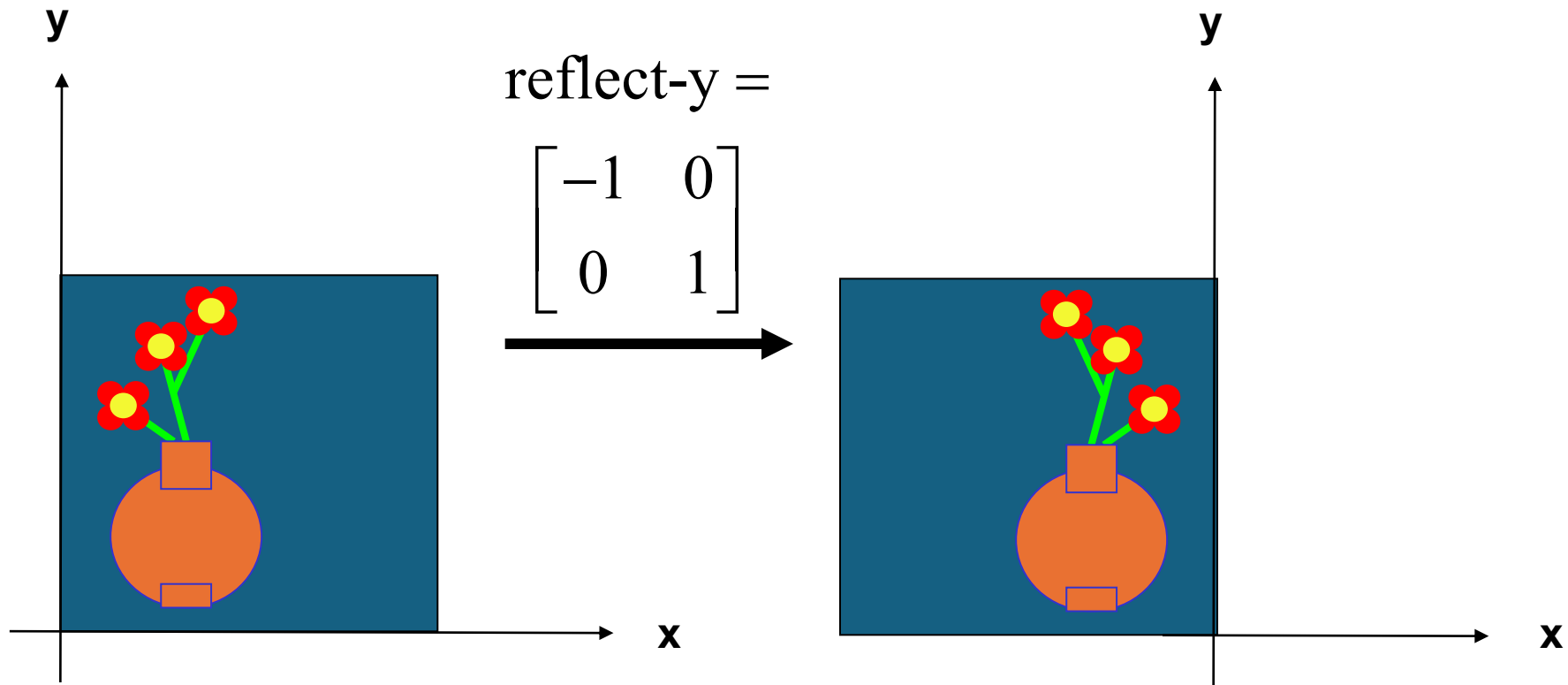
Rotation



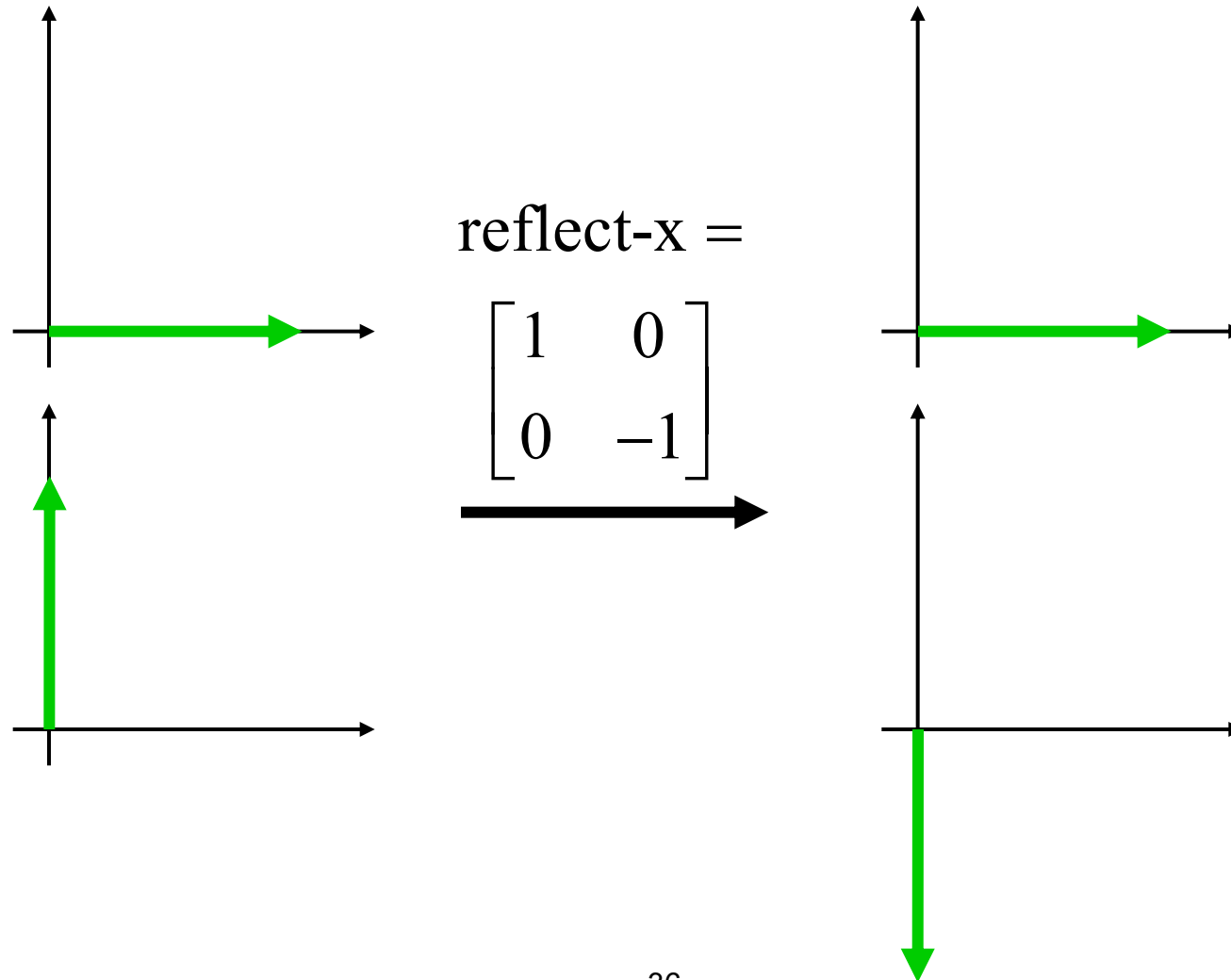
Reflection in y-axis



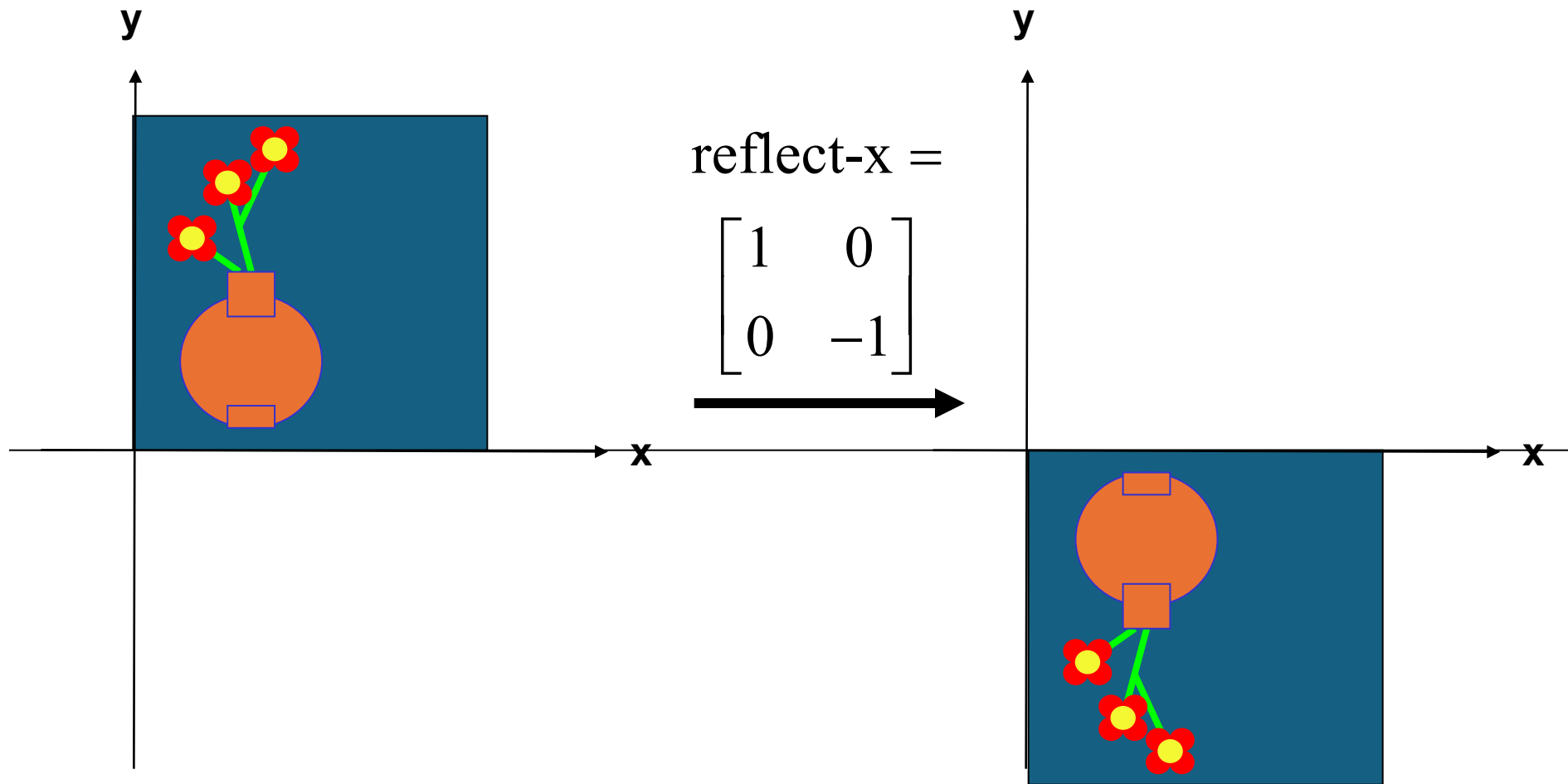
Reflection in y-axis



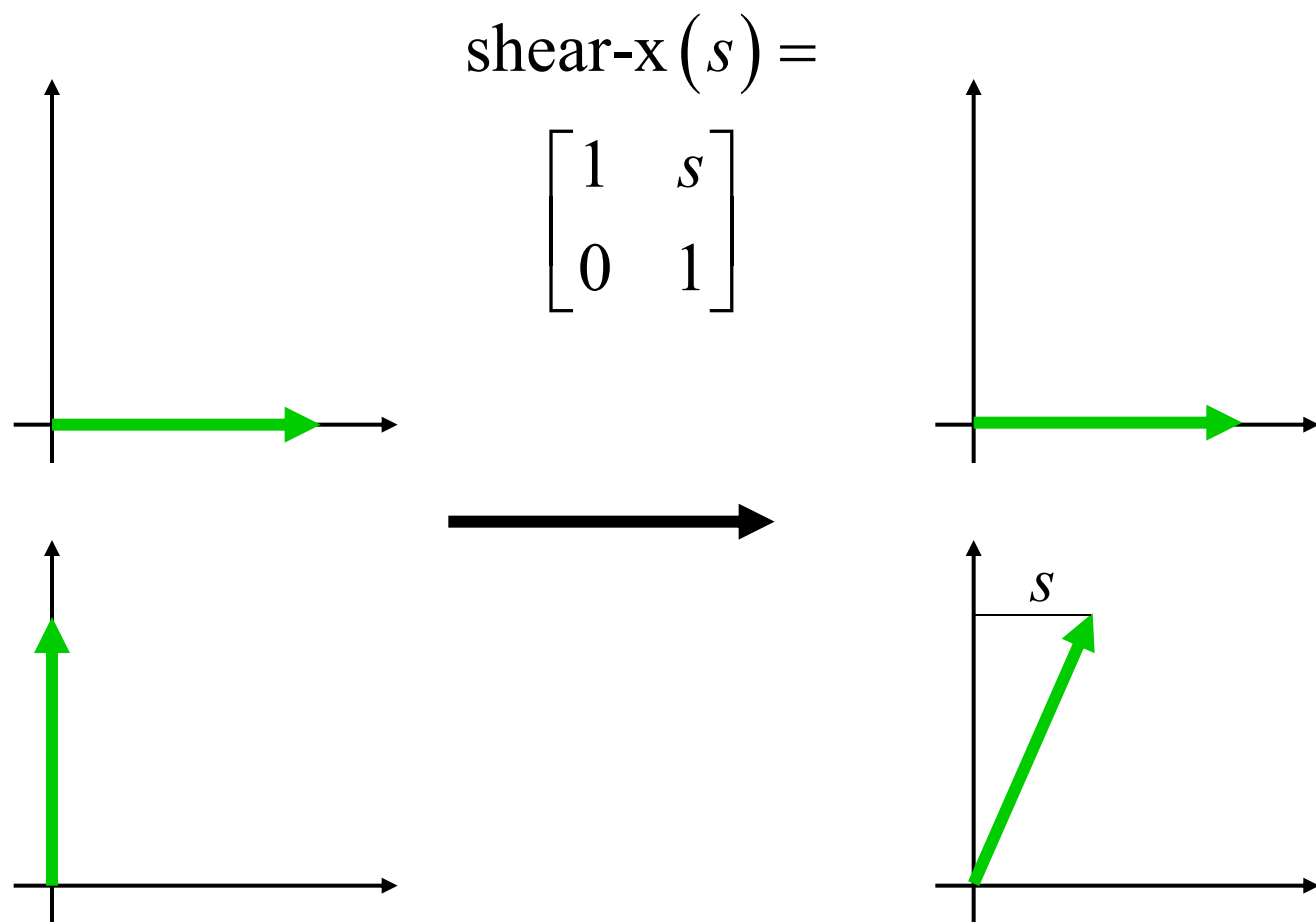
Reflection in x-axis



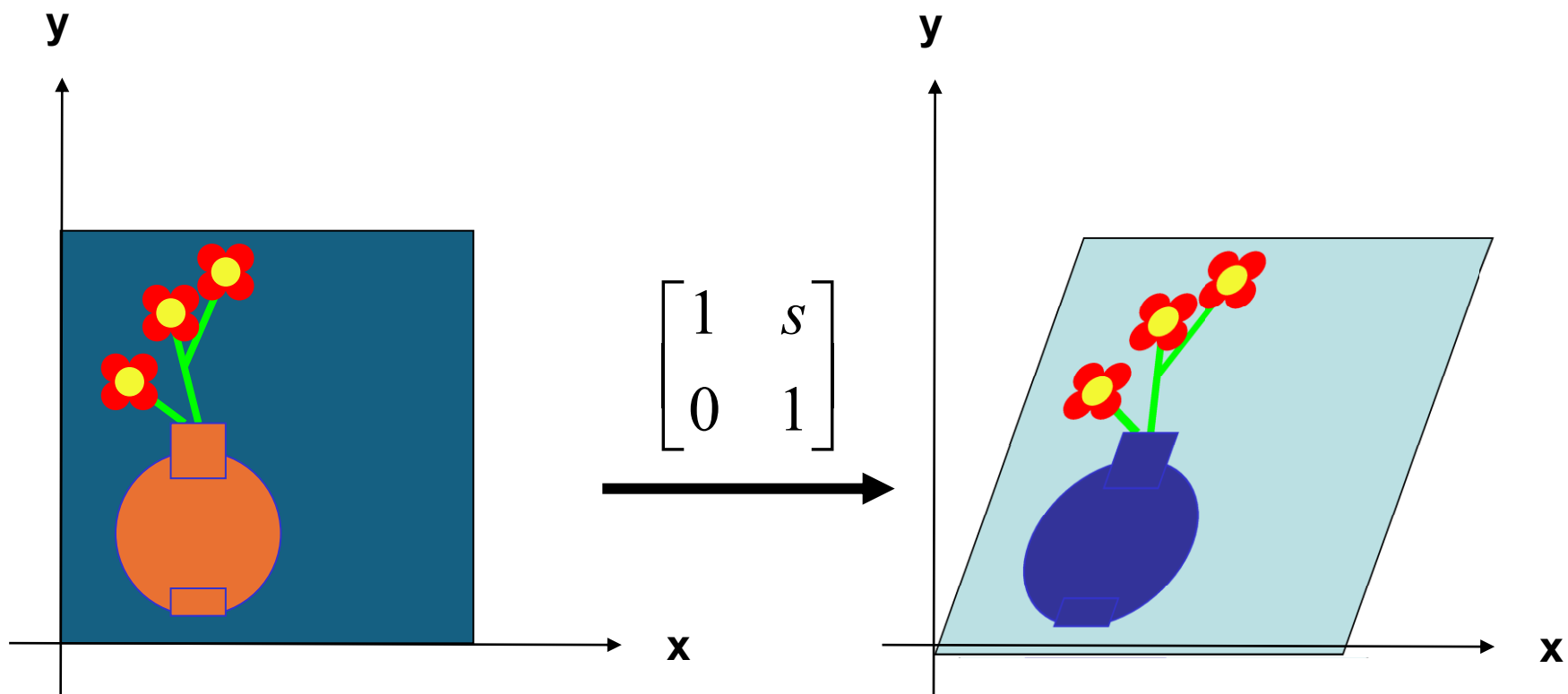
Reflection in x-axis



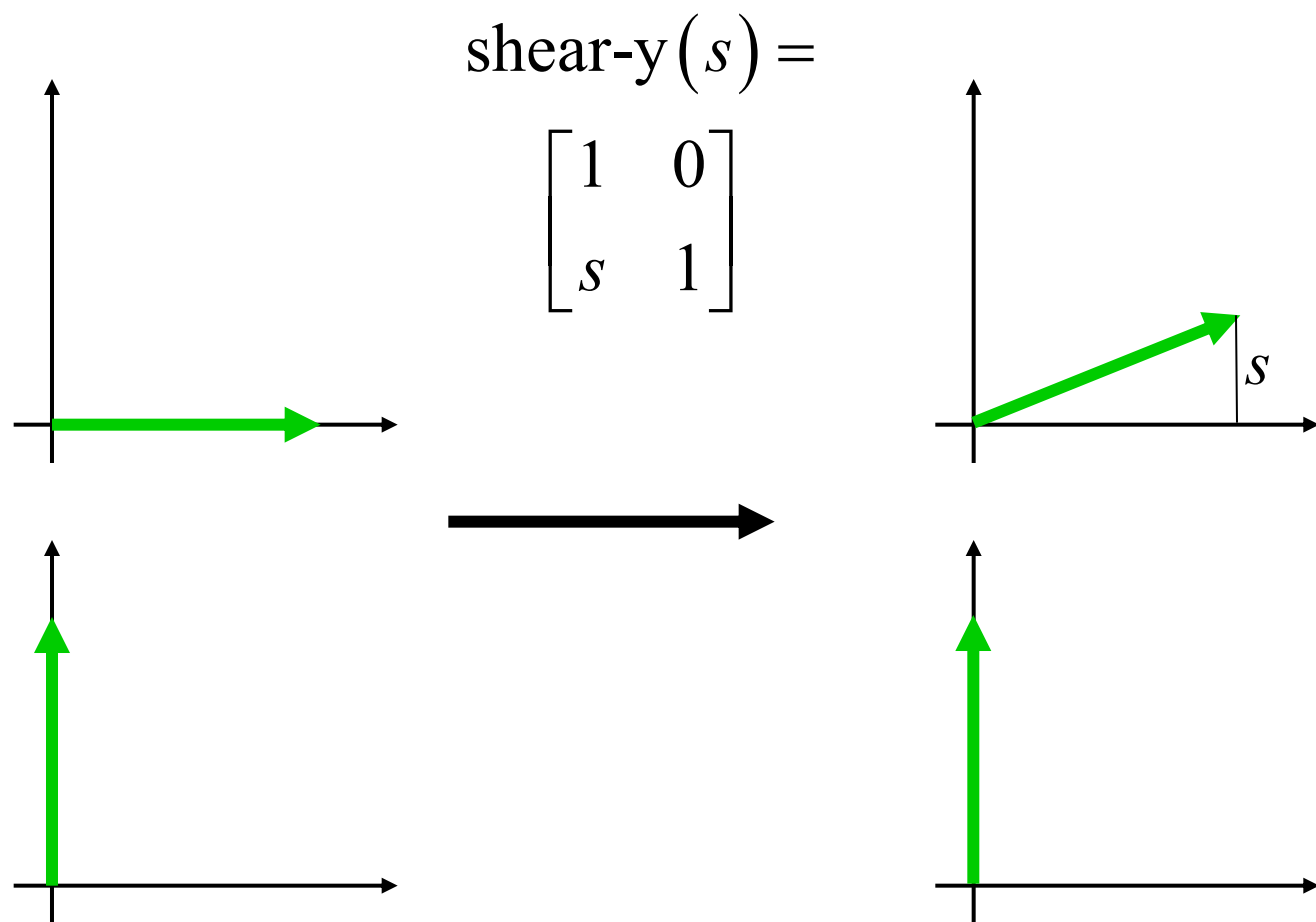
Shear-x



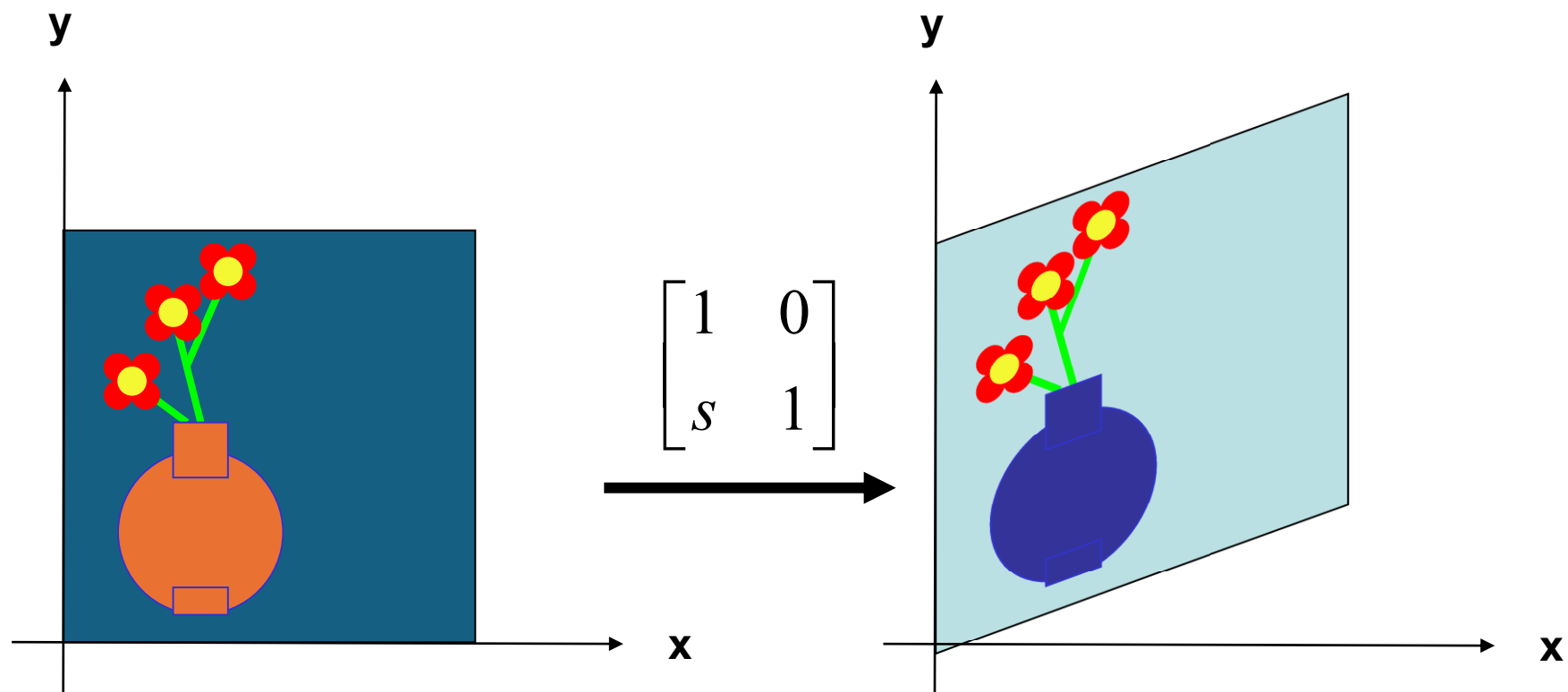
Shear x



Shear-y



Shear y



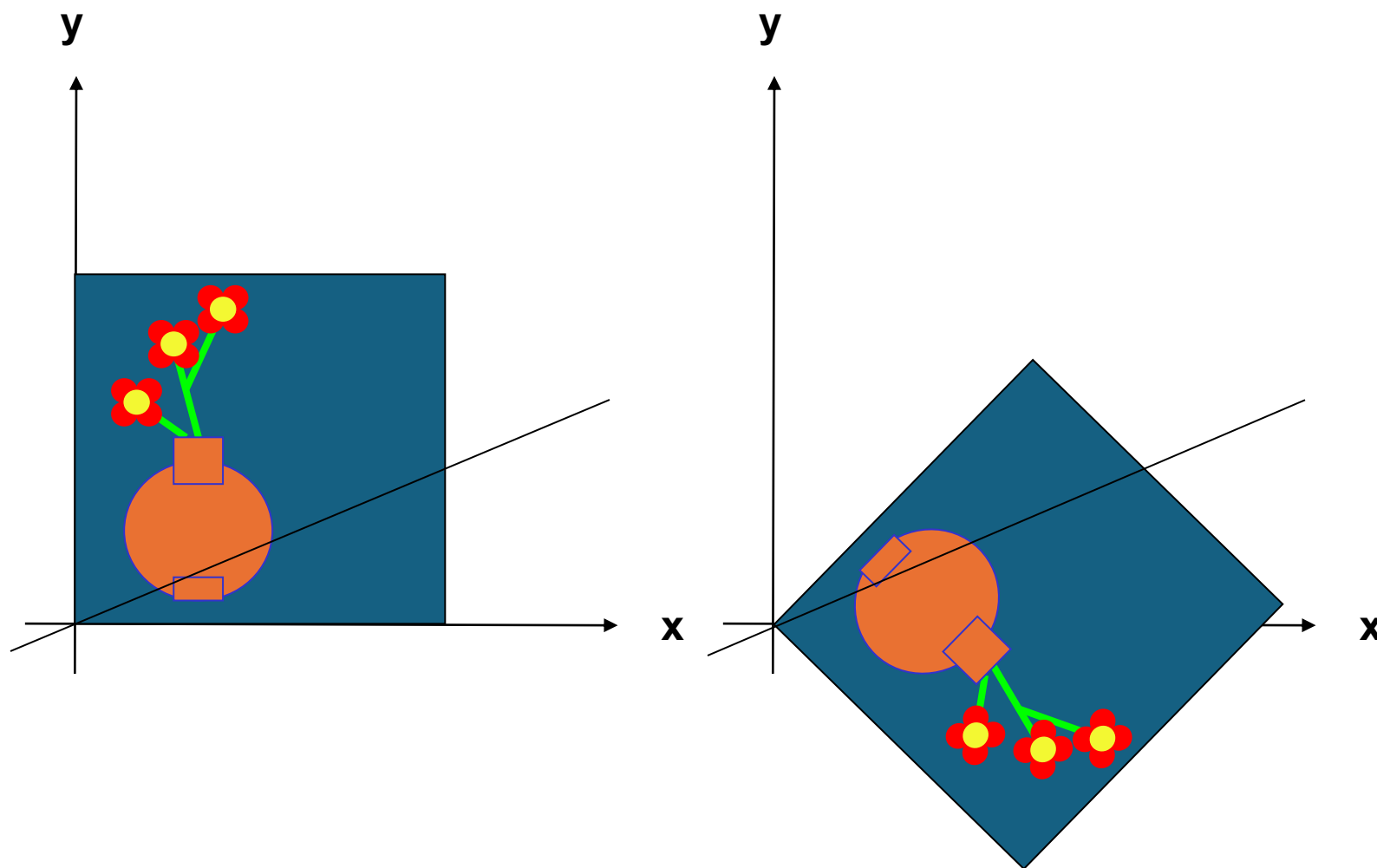
Linear Transformations

- Scale, Reflection, Rotation, and Shear are all linear transformations
- They satisfy: $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$
 - $\mathbf{u}$ and $\mathbf{v}$ are vectors
 - a and b are scalars
- If T is a linear transformation
 - $T((0, 0)) = (0, 0)$

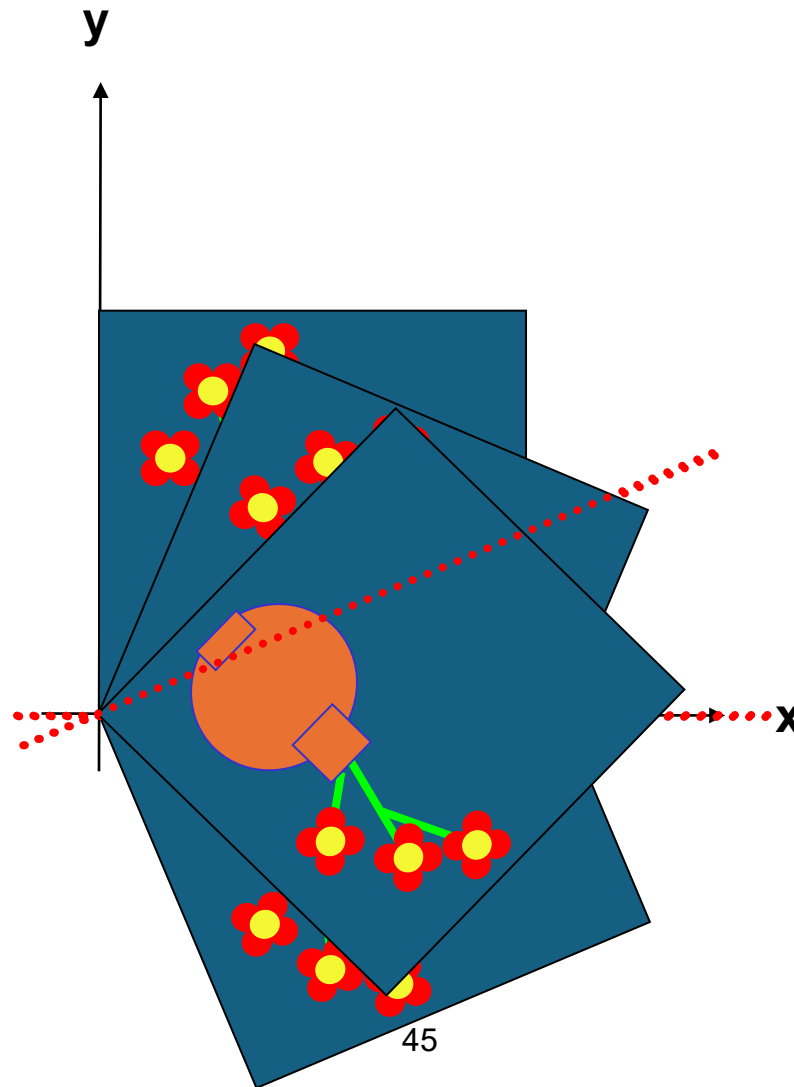
Composing Linear Transformations

- If T_1 and T_2 are transformations
 - $T_2 T_1(\mathbf{v}) =_{\text{def}} T_2(T_1(\mathbf{v}))$
- If T_1 and T_2 are linear and are represented by matrices M_1 and M_2
 - $T_2 T_1$ is represented by $M_2 M_1$
 - $T_2 T_1(\mathbf{v}) = T_2(T_1(\mathbf{v})) = (M_2 M_1)(\mathbf{v})$

Reflection About an Arbitrary Line (through the origin)



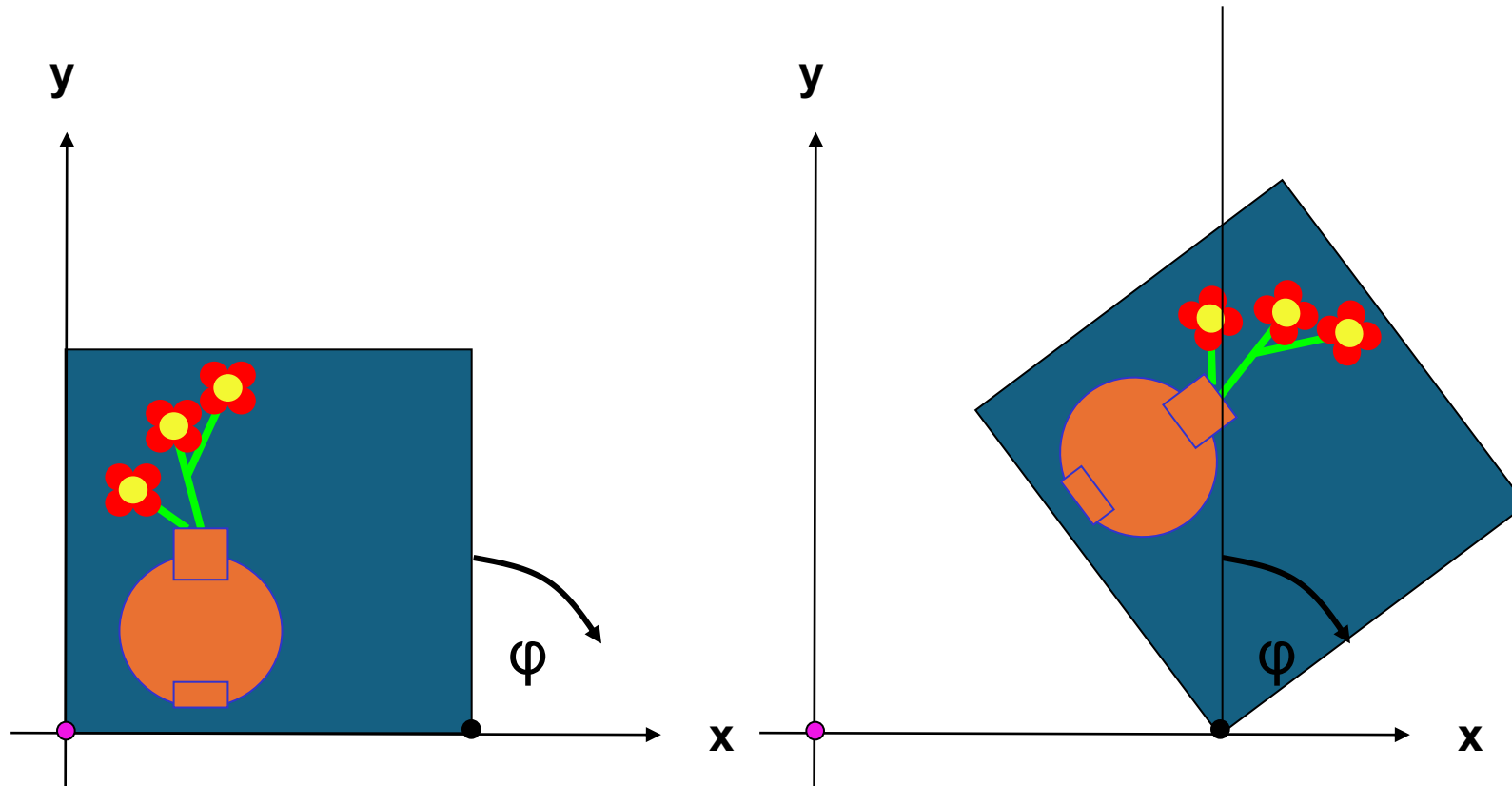
Reflection as a Composition



Decomposing Linear Transformations

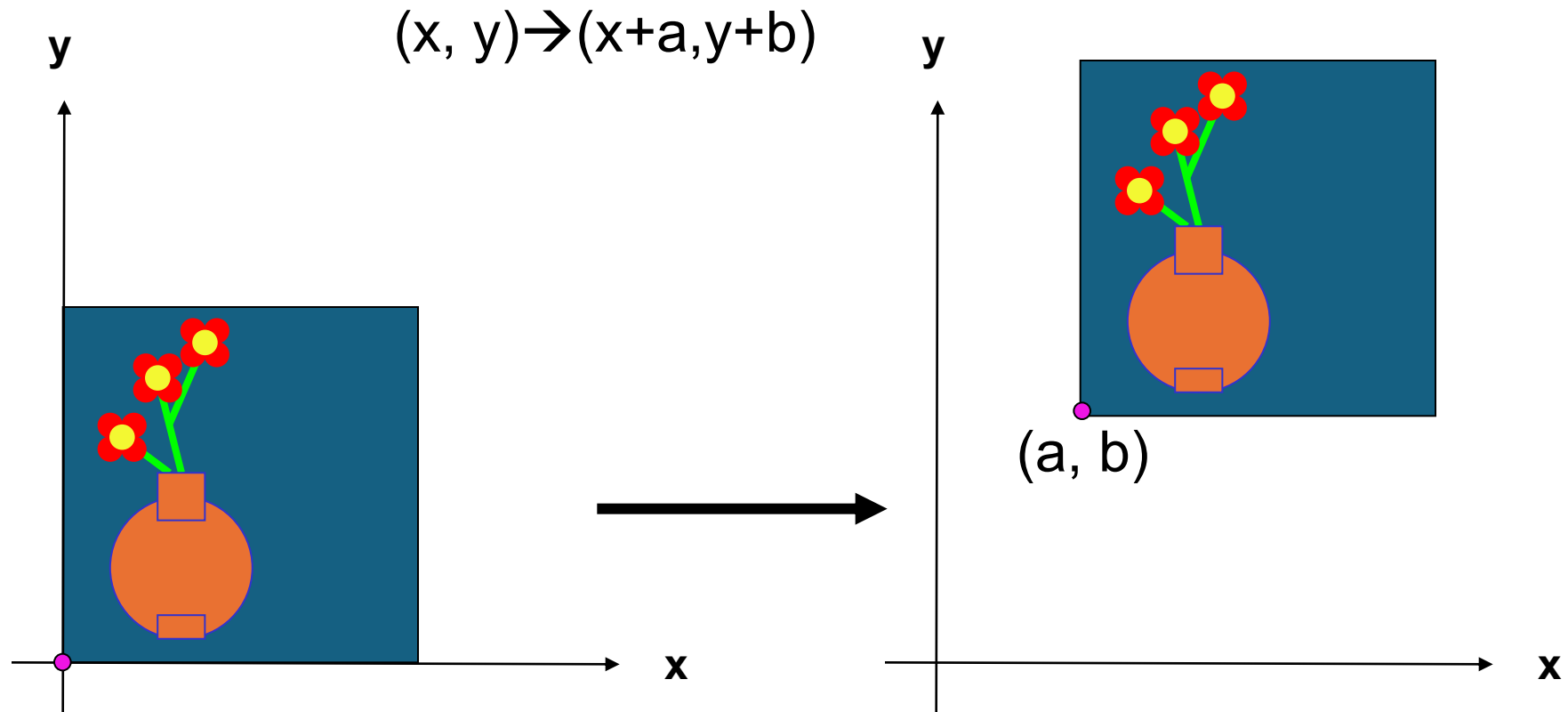
- Any 2D Linear Transformation can be decomposed into the product of a rotation, a scale, and a rotation if the scale can have negative numbers.
- $M = R_1 S R_2$

Rotation about an Arbitrary Point



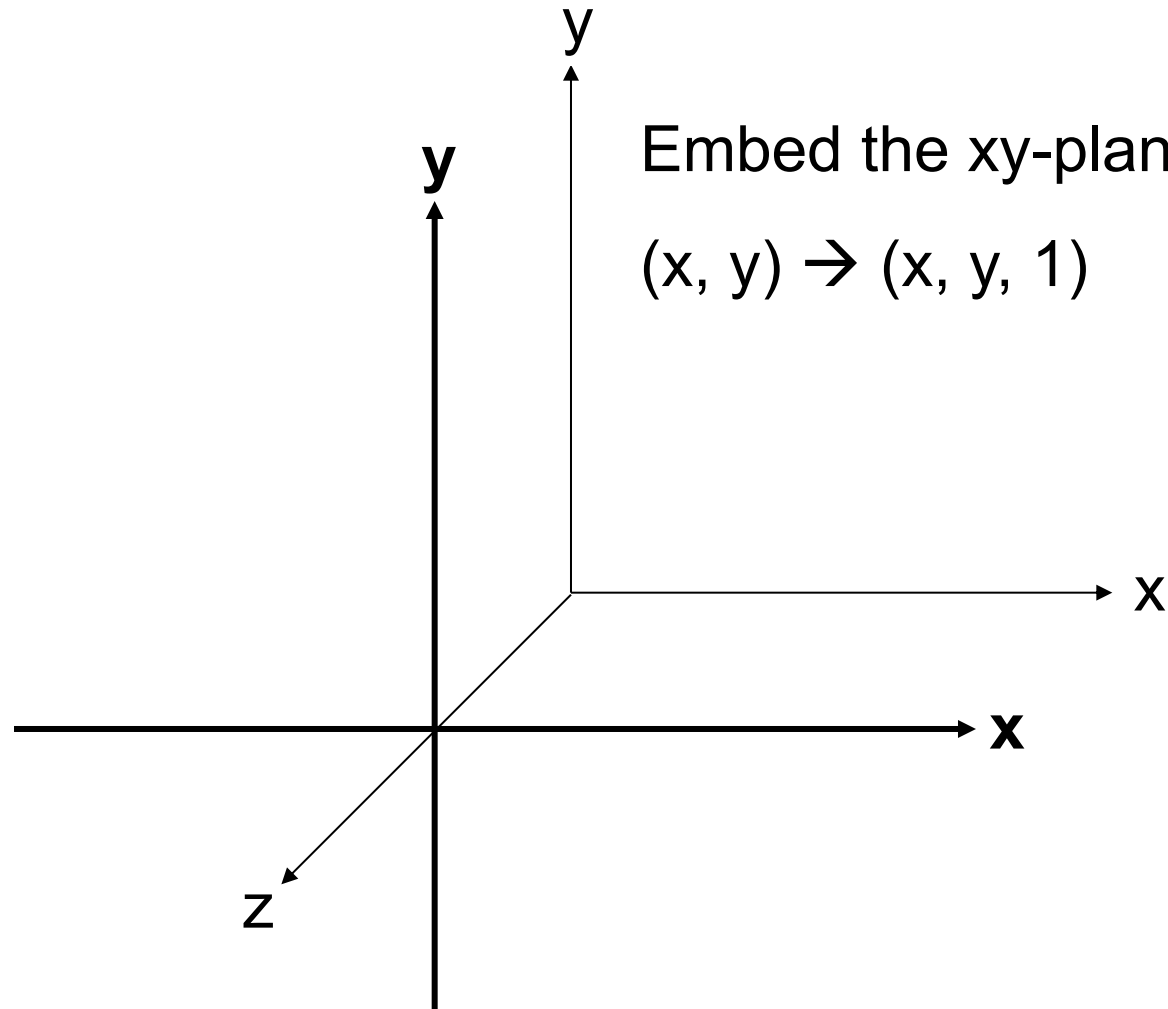
This is not a linear transformation. The origin moves.

Translation



This is not a linear transformation. The origin moves.

Homogeneous Coordinates



2D Linear Transformations as 3D Matrices

Any 2D linear transformation can be represented
by a 2x2 matrix

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

or a 3x3 matrix

$$\begin{bmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \\ 1 \end{bmatrix}$$

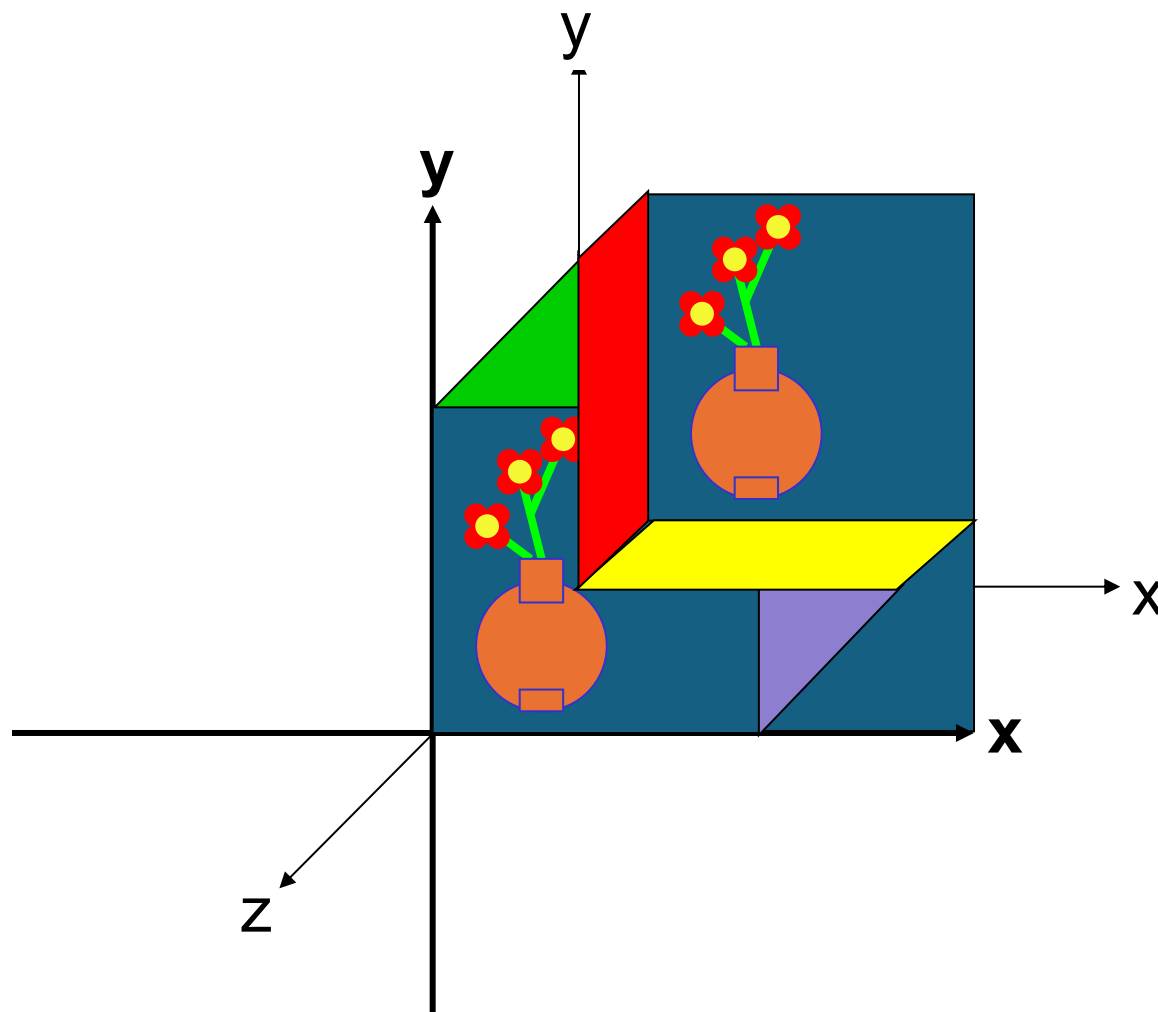
2D Linear Translations as 3D Matrices

Any 2D translation can be represented by a 3x3 matrix.

$$\begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + a \\ y + b \\ 1 \end{bmatrix}$$

This is a 3D shear that acts as a translation on the plane $z = 1$.

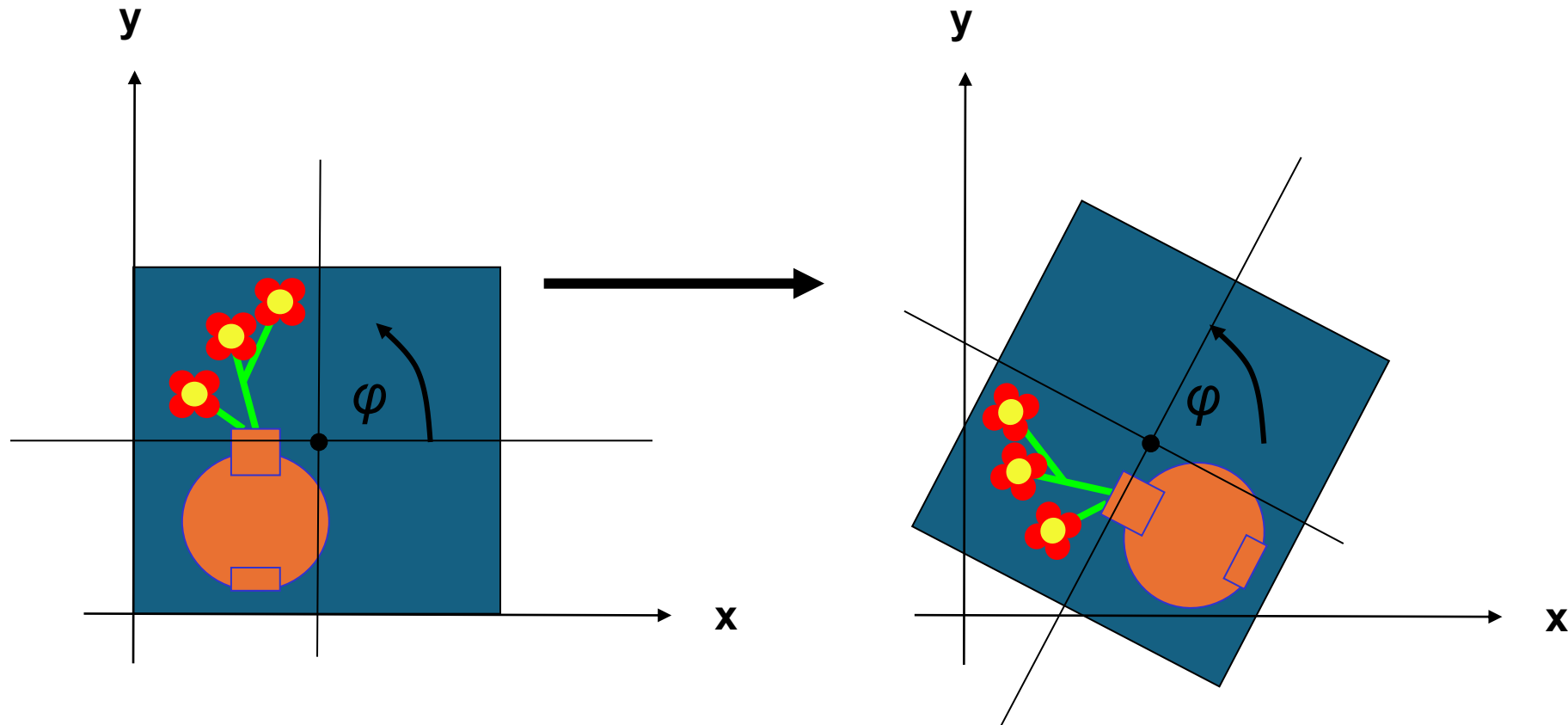
Translation as a Shear



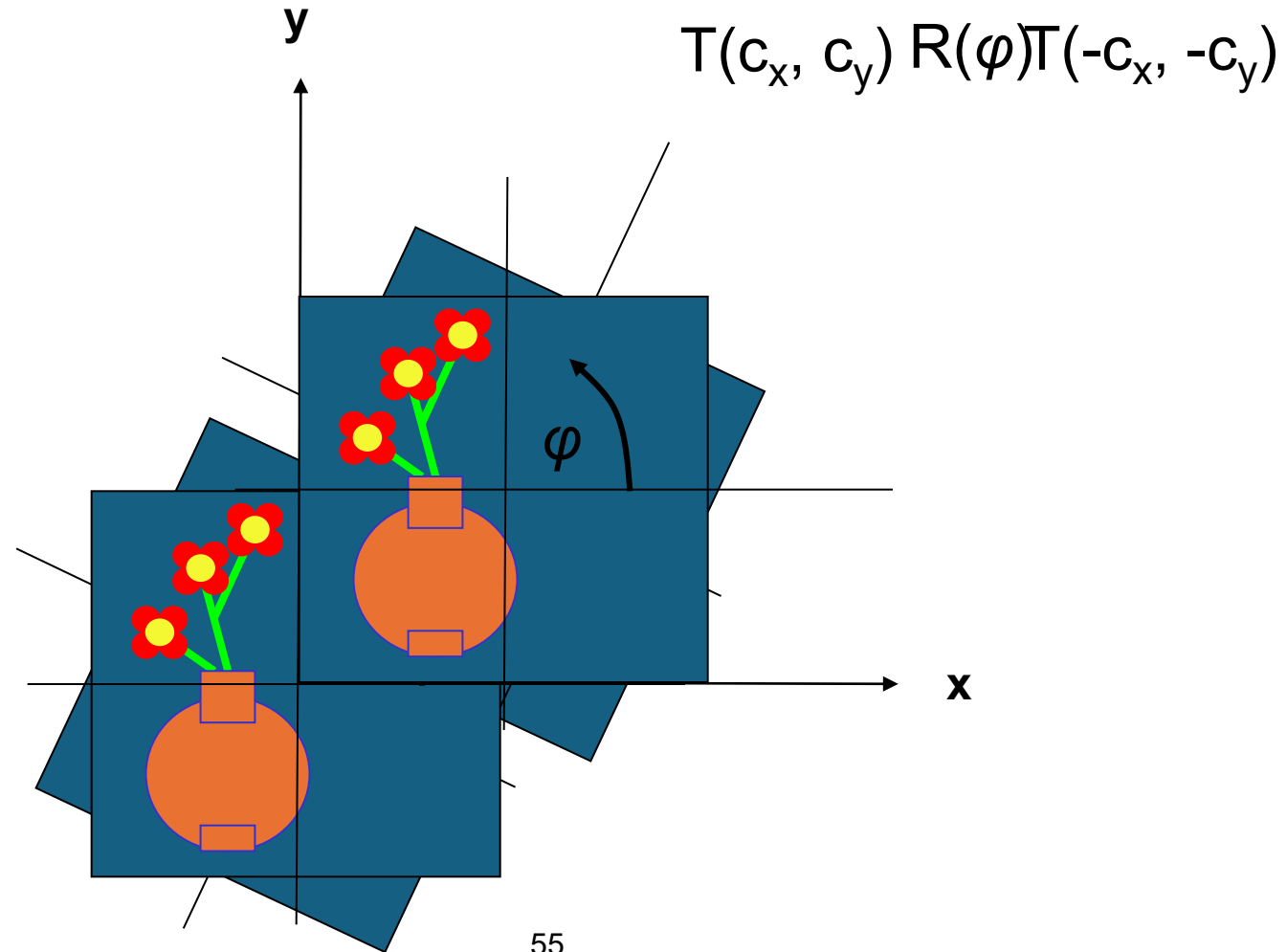
2D Affine Transformations

- An *affine transformation* is any transformation that preserves **co-linearity** (i.e., all points lying on a line initially still lie on a line after transformation) and **ratios of distances** (e.g., the midpoint of a line segment remains the midpoint after transformation).
- With homogeneous coordinates, we can represent all 2D affine transformations as 3D linear transformations.
- We can then use matrix multiplication to transform objects.

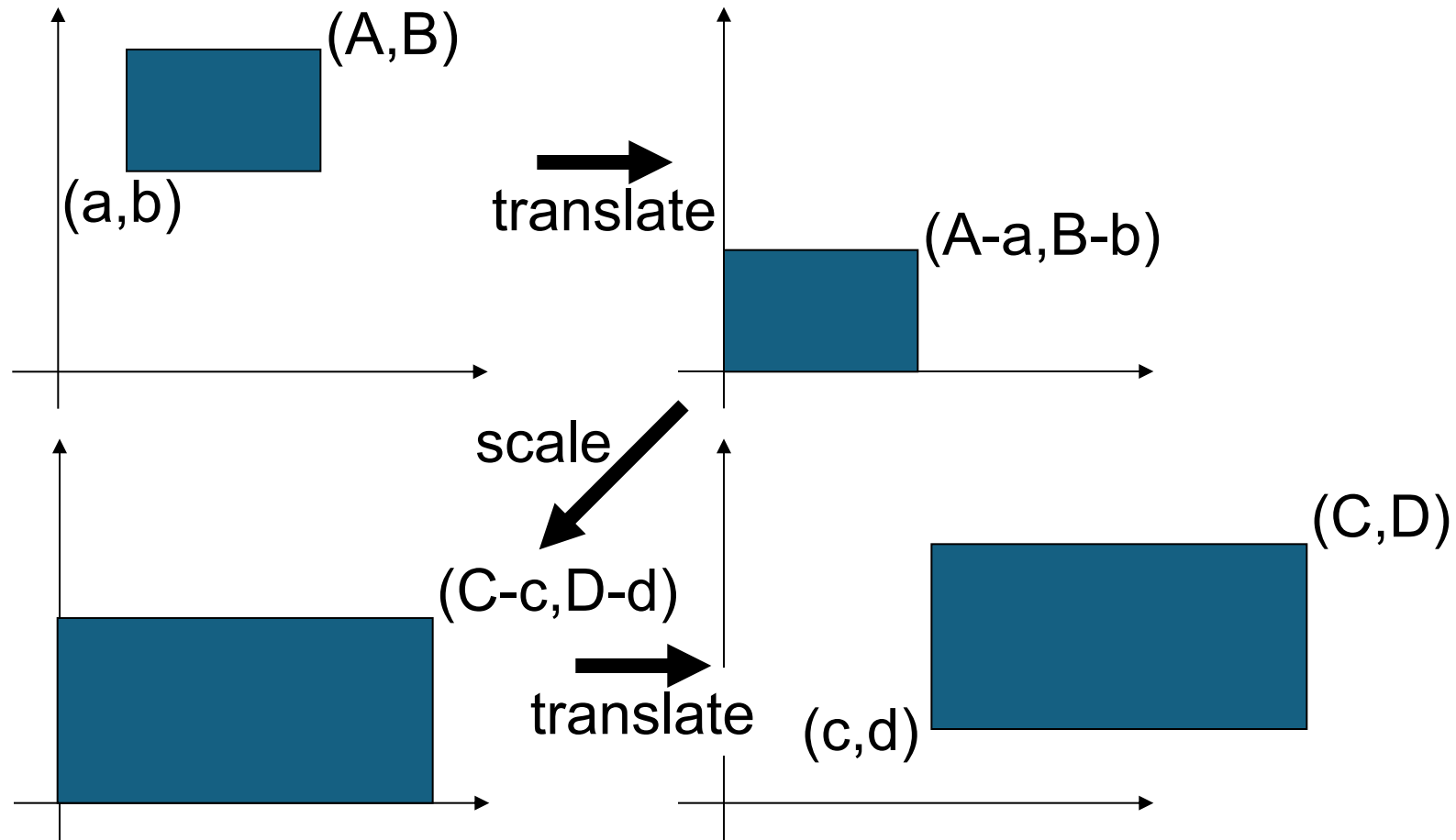
Rotation about an Arbitrary Point



Rotation about an Arbitrary Point



Windowing Transforms



3D Transformations

Remember:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \leftrightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

A 3D linear transformation can be represented by a 3x3 matrix.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \leftrightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Affine Transformations

$$\text{scale}(s_x, s_y, s_z) = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{translate}(t_x, t_y, t_z) = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Rotations

$$\text{rotate}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$\text{rotate}_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$\text{rotate}_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

QUANTUM VERSION

Column vector representative $\longrightarrow$ **one column matrix**

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

vector representatives in particular basis

**then
becomes**

$$\underline{y} = \underline{\underline{A}} \underline{x}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & & \\ \vdots & & & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$$

row vector $\longrightarrow$ **transpose of column vector**

$$\tilde{x} = (x_1, x_2 \cdots x_N)$$

$$\underline{y} = \underline{\underline{A}} \underline{x} \longrightarrow \tilde{y} = \tilde{x} \underline{\underline{A}} \quad \text{transpose}$$

$$\underline{y} = \underline{\underline{A}} \underline{x} \longrightarrow y^+ = x^+ \underline{\underline{A}}^+ \quad \text{Hermitian conjugate}$$

Change of Basis

orthonormal basis

$$\{|e^i\rangle\}$$

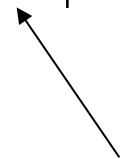
then

$$\langle e^i | e^j \rangle = \delta_{ij} \quad (i, j = 1, 2, \dots, N)$$

Superposition of $\{|e^i\rangle\}$  can form N new vectors
linearly independent

a new basis $\{|e^{i'}\rangle\}$

$$|e^{i'}\rangle = \sum_{k=1}^N u_{ik} |e^k\rangle \quad i = 1, 2, \dots, N$$

 complex numbers

New Basis is Orthonormal

$$\langle e^{j'} | e^{i'} \rangle = \delta_{ij}$$

if the matrix

$$\underline{\underline{U}} = (u_{ik})$$

coefficients in superposition

$$|e^{i'}\rangle = \sum_{k=1}^N u_{ik} |e^k\rangle \quad i = 1, 2, \dots, N$$

meets the condition

$$\underline{\underline{U}}^+ \underline{\underline{U}} = \underline{\underline{1}}$$

→ $\underline{\underline{U}}^{-1} = \underline{\underline{U}}^+ \quad \underline{\underline{U}} \text{ is unitary – Hermitian conjugate = inverse}$

Important result. The new basis $\{|e^{i'}\rangle\}$ will be orthonormal

if $\underline{\underline{U}}$, the transformation matrix, is unitary (see book

and Errata and Addenda, linear algebra book).

$$\underline{\underline{U}} \underline{\underline{U}}^+ = \underline{\underline{U}}^+ \underline{\underline{U}} = \underline{\underline{1}}$$

Unitary transformation

substitutes orthonormal basis $\{|e'\rangle\}$ for orthonormal basis $\{|e\rangle\}$.

Vector $|x\rangle$

$$|x\rangle = \sum_i x_i |e^i\rangle$$

vector – line in space (may be high dimensionality
abstract space)

$$|x\rangle = \sum_i x'_i |e^{i'}\rangle$$

written in terms of two basis sets

$|x\rangle$ Same vector – different basis.

The unitary transformation $\underline{\underline{U}}$ can be used to change a vector representative of $|x\rangle$ in one orthonormal basis set to its vector representative in another orthonormal basis set.

x – vector rep. in unprimed basis

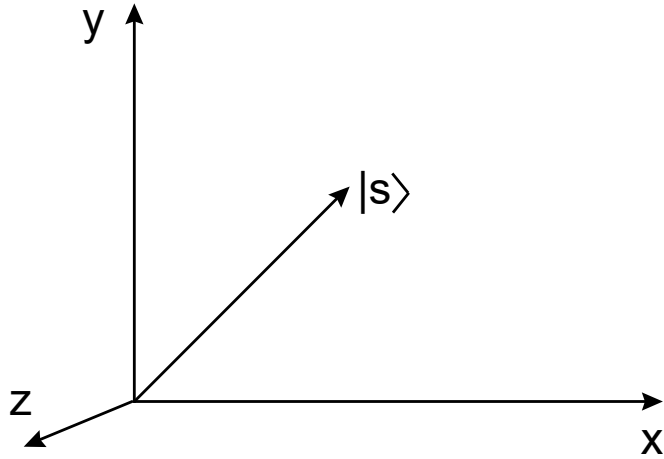
x' – vector rep. in primed basis

$$x' = \underline{\underline{U}} x \quad \text{change from unprimed to primed basis}$$

$$x = \underline{\underline{U}}^+ x' \quad \text{change from primed to unprimed basis}$$

Example

Consider basis $\{\hat{x}, \hat{y}, \hat{z}\}$



Vector $|s\rangle$ - line in real space.

In terms of basis $|s\rangle = 7\hat{x} + 7\hat{y} + 1\hat{z}$

Vector representative in basis $\{\hat{x}, \hat{y}, \hat{z}\}$

$$s = \begin{bmatrix} 7 \\ 7 \\ 1 \end{bmatrix}$$

Change basis by rotating axis system 45° around $\hat{z}$.

Can find the new representative of $|s\rangle, s'$

$$s' = \underline{\underline{U}} s$$

$\underline{\underline{U}}$ is rotation matrix

$$\underline{\underline{U}} = \begin{pmatrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{matrix} \end{pmatrix}$$

For 45° rotation around z

$$\underline{\underline{U}} = \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then

$$s' = \begin{pmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 7\sqrt{2} \\ 0 \\ 1 \end{pmatrix}$$

$$s' = \begin{bmatrix} 7\sqrt{2} \\ 0 \\ 1 \end{bmatrix} \quad \text{vector representative of } |s\rangle \text{ in basis } \{|e'\rangle\}$$

Same vector but in new basis.

Properties unchanged.

Example – length of vector $(\langle s|s\rangle)^{1/2}$

$$[\langle s|s\rangle]^{1/2} = (s^* \cdot s)^{1/2} = (49 + 49 + 1)^{1/2} = (99)^{1/2}$$

$$[\langle s|s\rangle]^{1/2} = (s'^* \cdot s')^{1/2} = (2 \times 49 + 0 + 1)^{1/2} = (99)^{1/2}$$